ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 8

MATHEMATICS

PAPER 1 · PRACTICE B · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central Grade 8 Mathematics paper 1 · practice b.

Written by Academic Central for practice, following the CAPS curriculum for Grade 8. It is not a Department of Basic Education marking guideline.

How to mark: each ✓ is one mark. Give method marks for correct working even if the final answer is wrong (consistent accuracy). Accept other correct answers.

QUESTION 1

1 1.1 C ✓ 1.2 A ✓ 1.3 D ✓ 1.4 B ✓ 1.5 D ✓ 1.6 B ✓ 1.7 C ✓ 1.8 A ✓ 1.9 B ✓ 1.10 D ✓ [10]

QUESTION 2

2.1 $2 + 3 + 5 = 10$ parts; R2 400 ÷ 10 = R240 per part (3)
Ayanda R480 ✓ Bongi R720 ✓ Chris R1 200 ✓

2.2 (a) $360 \div 100 \times 7{,}5$ ✓ $= 27\ \ell$ ✓ (b) $27 \times$ R23,40 = R631,80 ✓ (3)

2.3 (a) 15% × R600 = R90 ✓ → R690 ✓ (4)
(b) 20% × R690 = R138 ✓ → R690 − R138 = R552 ✓

[10]

QUESTION 3

3.1 (a) $36 \div (-4) + (-27)$ ✓ $= -9 - 27$ ✓ $= -36$ ✓ (5)
(b) $12 - (-5)$ ✓ $= 17$ ✓

3.2 (a) $a^8 \div a^4$ ✓ $= a^4$ ✓ (b) $9x^4y^2$ ✓✓ (c) $8x^3$ ✓ (5)

[10]

QUESTION 4

4.1 (a) $^{13}/_4 - {}^{11}/_6$ ✓ $= {}^{39}/_{12} - {}^{22}/_{12} = {}^{17}/_{12} = 1\,{}^5/_{12}$ ✓✓ (7)
(b) ${}^8/_3 \div {}^4/_3 = {}^8/_3 \times {}^3/_4$ ✓ $= 2$ ✓
(c) ${}^5/_6 + {}^3/_6$ ✓ $= {}^8/_6 = 1\,{}^1/_3$ ✓

4.2 3,142 ✓ (1)

[8]

QUESTION 5

5.1.1 $T_n = 2n + 1$ ✓✓ (2)

5.1.2 $2(50) + 1 = 101$ ✓ (1)

5.1.3 $2n + 1 = 81 \rightarrow 2n = 80$ ✓ → Figure 40 ✓ (2)

5.2 (a) $8x - 12 - 3x + 15$ ✓✓ $= 5x + 3$ ✓ (8)
(b) $x^3 - 2x^2$ ✓ $+ 3x$ ✓
(c) $3x^2y$ ✓ $- 1$ ✓
(d) $4x^4$ ✓

5.3 (a) $2x = 14$ ✓ $\rightarrow x = 7$ ✓ (6)
(b) $x - 1 = 12$ ✓ $\rightarrow x = 13$ ✓
(c) $x^2 = 25$ ✓ $\rightarrow x = 5$ or $x = -5$ ✓

[19]

QUESTION 6

6.1 $3x + 10° = 2x + 30°$ (vertically opposite angles) ✓ $\rightarrow x = 20°$ ✓ (4)
$\angle AOC = 3(20°) + 10° = 70°$ ✓ $\angle COB = 180° - 70° = 110°$ (adjacent angles on a straight line) ✓

6.2 (a) $\angle ACB = 180° - 115° = 65°$ (adjacent angles on a straight line) ✓ (4)
(b) $\angle B = \angle ACB = 65°$ (base angles of an isosceles triangle, AB = AC) ✓✓
(c) $\angle A = 180° - 65° - 65° = 50°$ (sum of the angles of a triangle) ✓

6.3 $x + 2x = 180°$ (co-interior angles, AB ∥ CD) ✓ $\rightarrow 3x = 180°$ ✓ $\rightarrow x = 60°$ ✓ (3)

6.4 $^{EF}/_{BC} = {}^{DE}/_{AB} \rightarrow EF = 8 \times {}^{9}/_{6}$ ✓ $= 12$ cm ✓ (2)

6.5 Square ✓ (1)

[14]

QUESTION 7

7.1 $d^2 = 80^2 + 60^2 = 10\,000$ ✓ $\rightarrow d = 100$ cm ✓ (2)

7.2 $x^2 = 13^2 - 5^2 = 144$ ✓ $\rightarrow x = 12$ cm ✓ (2)

7.3 (a) $A = 3{,}14 \times 10^2$ ✓ $= 314$ cm^2 ✓ (4)
(b) ½ × 2 × 3,14 × 10 + 20 ✓ $= 31{,}4 + 20 = 51{,}4$ cm ✓

7.4 (a) $V = 60 \times 30 \times 40$ ✓ $= 72\,000$ cm^3 ✓ $= 72\ \ell$ ✓ (6)
(b) Base $60 \times 30 = 1\,800$; front and back $2(60 \times 40) = 4\,800$; sides $2(30 \times 40) = 2\,400$ ✓✓
Total $= 9\,000$ cm^2 ✓

[14]

QUESTION 8

8.1.1 A'$(-1\,;-1)$ ✓ B'$(-4\,;-1)$ ✓ C'$(-1\,;-3)$ ✓ (3)

8.1.2 $(-1\,;3)$ ✓ (1)

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8.1.3 $(12\ ;\ 3)$ ✓ (1)

[5]

QUESTION 9

9.1.1 2 books ✓ (1)

9.1.2 $0(4) + 1(7) + 2(9) + 3(6) + 4(4)$ ✓ $= 0 + 7 + 18 + 18 + 16 = 59$ books ✓ (2)

9.1.3 $59 \div 30 = 1{,}97$ books ✓ (1)

9.1.4 The median lies between the 15th and 16th values ✓; the first 11 learners read 0 or 1 book and learners 12 to 20 read 2 books, so the median is 2 ✓ (2)

9.1.5 $4 - 0 = 4$ ✓ (1)

9.1.6 $^{6+4}/_{30} = {}^{10}/_{30}$ ✓ $= {}^{1}/_{3}$ ✓ (2)

9.1.7 A bar graph ✓ (1)

[10]

TOTAL: 100

END OF MARKING GUIDELINE